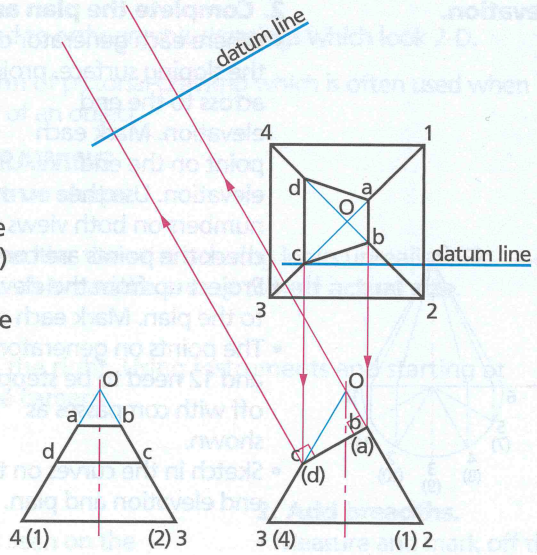
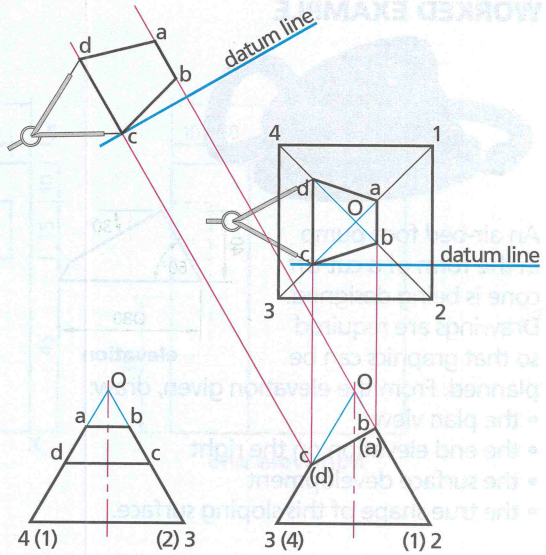
6. Draw the true shape of the top.

- Choose the view which shows the sloping surface as a single line (in this example it is the elevation).
- Project at 90° from this surface (this gives the true length of the surface).
- Draw a **datum line** at 90° to the projection lines. (A datum line is a line from which measurements are taken.)
- Add a horizontal datum line to the plan. This datum line can be anywhere on the plan but it is normal to pass it through the lowest corner of the surface or through the centre of the surface.



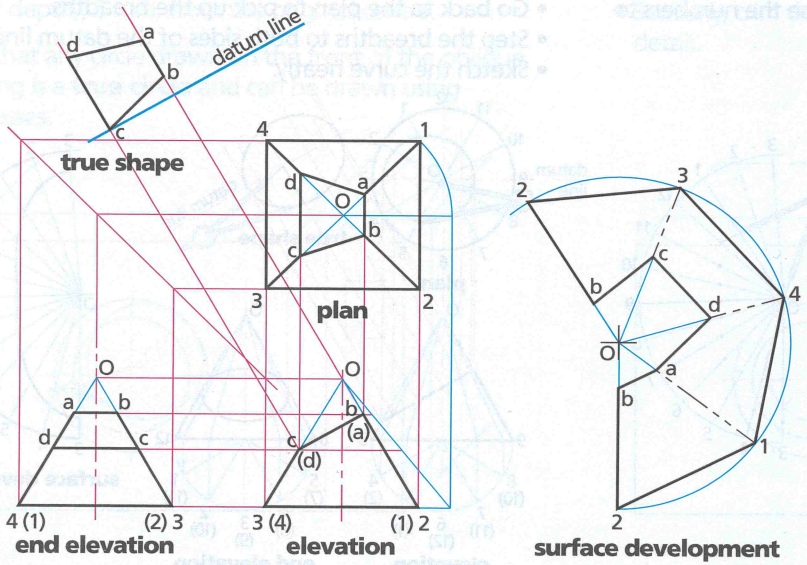
7. Complete the true shape.

- Take the breadths above the datum line on the plan.
- Step them off above the other datum line.
- Mark and label points a, b, c and d.
- Join them to find the true shape of the top surface.



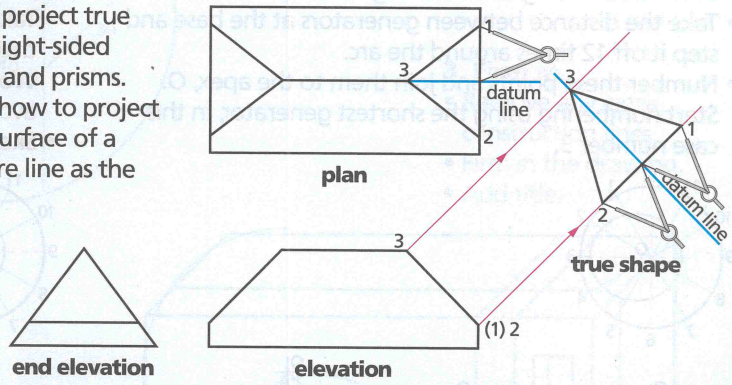
8. The finished drawing.

Here is the complete drawing showing elevation, plan, end elevation, surface development and true shape.



WORKED EXAMPLE 2

You can be asked to project true shapes from any straight-sided forms, e.g. pyramids and prisms. This example shows how to project the true shape of a surface of a prism using the centre line as the datum.



1. Choose the view which shows the surface as a single line.
2. Project at 90° from the surface.
3. Draw a datum line at 90° to the projection lines.
4. Add a horizontal datum line to the plan. In this example, the datum line passes through the centre of the surface.
5. Take the breadths above the datum line on the plan.
6. Step them off on both sides of the other datum line.
7. Mark and label points 1, 2 and 3.
8. Join them to find the true shape of the sloping surface.

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